

# Linear Transformations II

Jin-Ren Ryan Wang

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# Contents

# 1 Isomorphism of vector spaces

## 1.1 Definition

**Definition 5.1.1. (inverse function)** A function  $f : A \rightarrow B$  is called *bijective* if  $f$  is both injective and surjective. If  $f : A \rightarrow B$  is bijective, then its inverse function  $f^{-1} : B \rightarrow A$  exists and satisfies  $f^{-1}(f(a)) = a$ ,  $\forall a \in A$  and  $f(f^{-1}(b)) = b$ ,  $\forall b \in B$ .

**Definition 5.1.2.** Let  $V$  and  $W$  be vector spaces over a field  $F$ .

1.  $f : V \rightarrow W$  is called a *(linear) isomorphism from  $V$  onto  $W$*  if  $f$  is a bijective linear map.
2. We say that  $V$  and  $W$  are *isomorphic as vector spaces* if there exists a linear isomorphism  $f : V \rightarrow W$ .

## 1.2 Theorem

**Theorem 5.1.1.** Let  $f : V \rightarrow W$  be a linear map of vector spaces. If  $f$  is bijective (and hence  $f^{-1}$  exists), then  $f^{-1} : W \rightarrow V$  is also linear.

*Proof.* Since  $f$  is surjective, every  $\mathbf{w} \in W$  can be written as  $\mathbf{w} = f(\mathbf{v})$  for some  $\mathbf{v} \in V$ . We verify the two properties of linearity.

### 1. Additivity.

Let  $\mathbf{w}_1 = f(\mathbf{v}_1)$ ,  $\mathbf{w}_2 = f(\mathbf{v}_2)$ , where  $\mathbf{v}_1, \mathbf{v}_2 \in V$ . Then

$$\begin{aligned} f^{-1}(\mathbf{w}_1 + \mathbf{w}_2) &= f^{-1}(f(\mathbf{v}_1) + f(\mathbf{v}_2)) \\ &= f^{-1}(f(\mathbf{v}_1 + \mathbf{v}_2)) \\ &= \mathbf{v}_1 + \mathbf{v}_2 \\ &= f^{-1}(\mathbf{w}_1) + f^{-1}(\mathbf{w}_2). \end{aligned}$$

### 2. Homogeneity.

Let  $\lambda \in F$ ,  $\mathbf{w} \in W$ , so  $\mathbf{w} = f(\mathbf{v})$  for some  $\mathbf{v} \in V$ . Then

$$\begin{aligned} f^{-1}(\lambda \mathbf{w}) &= f^{-1}(\lambda f(\mathbf{v})) \\ &= f^{-1}(f(\lambda \mathbf{v})) \\ &= \lambda \mathbf{v} \\ &= \lambda f^{-1}(\mathbf{w}). \end{aligned}$$

This shows  $f^{-1} : W \rightarrow V$  is also linear. □

**Theorem 5.1.2.** Let  $V$  be a vector space over a field  $F$  of dimension  $n$ . Let  $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$  be a basis of  $V$ . Then the function  $\phi_S : V \rightarrow F^n$  defined by

$$\mathbf{v} \mapsto [\mathbf{v}]_S.$$

is a linear isomorphism.

*Proof.* Write

$$\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad \mathbf{v}' = \sum_{i=1}^n \alpha'_i \mathbf{v}_i.$$

We prove the statement in three steps.

1.  $\phi_S$  is linear.

By definition,  $[\mathbf{v}]_S = (\alpha_1, \alpha_2, \dots, \alpha_n)$ .

Hence

$$\begin{aligned} \phi_S(\mathbf{v} + \mathbf{v}') &= \phi_S\left(\sum_{i=1}^n (\alpha_i + \alpha'_i) \mathbf{v}_i\right) \\ &= (\alpha_1 + \alpha'_1, \dots, \alpha_n + \alpha'_n) \\ &= (\alpha_1, \dots, \alpha_n) + (\alpha'_1, \dots, \alpha'_n) \\ &= \phi_S(\mathbf{v}) + \phi_S(\mathbf{v}'). \end{aligned}$$

Likewise, for any  $\lambda \in F$ ,

$$\begin{aligned} \phi_S(\lambda \mathbf{v}) &= \phi_S\left(\sum_{i=1}^n (\lambda \alpha_i) \mathbf{v}_i\right) \\ &= (\lambda \alpha_1, \dots, \lambda \alpha_n) \\ &= \lambda(\alpha_1, \dots, \alpha_n) \\ &= \lambda \phi_S(\mathbf{v}). \end{aligned}$$

Therefore,  $\phi_S$  is linear.

2.  $\phi_S$  is injective.

Let  $\mathbf{v} \in \ker(\phi_S) \implies \phi_S(\mathbf{v}) = (0, \dots, 0) \implies \mathbf{v} = \sum_{i=1}^n 0 \mathbf{v}_i = \mathbf{0}_V$ .

Thus

$$\ker(\phi_S) = \{\mathbf{0}\},$$

and therefore  $\phi_S$  is injective.

3.  $\boxed{\phi_S \text{ is surjective.}}$

By the Rank–Nullity Theorem,

$$\text{rank}(\phi_S) + \text{nullity}(\phi_S) = \underbrace{\dim_F(V)}_n.$$

Since  $\phi_S$  is injective,

$$\text{nullity}(\phi_S) = 0.$$

Hence

$$\text{rank}(\phi_S) = n = \dim_F(V).$$

Because  $\dim_F(F^n) = n$ , we conclude that  $\text{im}(\phi_S) = F^n$ . Therefore,  $\phi_S$  is surjective.

Hence  $\phi_S$  is a linear isomorphism.  $\square$

**Theorem 5.1.3.** Let  $V$  and  $W$  be two finite-dimensional vector spaces over a field  $F$ . Then  $V$  and  $W$  are isomorphic as vector spaces if and only if  $\dim_F(V) = \dim_F(W)$ .

*Proof.* We prove both directions.

( $\Rightarrow$ ) Suppose  $V$  and  $W$  are isomorphic. By definition,  $\exists$  a linear isomorphism  $f : V \rightarrow W$

Since  $f$  is injective  $\iff \text{nullity}(f) = 0$ , and  $f$  is surjective  $\iff \text{rank}(f) = \dim_F(W)$ .

By the Rank–Nullity Theorem,  $\text{rank}(f) + \text{nullity}(f) = \dim_F(V)$ .

Hence  $\dim_F(W) + 0 = \dim_F(V)$ , so  $\dim_F(V) = \dim_F(W)$ .

( $\Leftarrow$ ) Suppose  $\dim_F(V) = \dim_F(W) = n$ . Fix a basis  $S$  of  $V$ , and a basis  $S'$  of  $W$ .

By Theorem 5.1.2, the coordinate maps

$$\phi_S : V \rightarrow F^n, \quad \phi_{S'} : W \rightarrow F^n$$

are linear isomorphisms.

$$\text{Now, } V \xrightarrow{\phi_S} F^n \xrightarrow{\phi_{S'}^{-1}} W$$

Therefore, their composition  $\phi_{S'}^{-1} \circ \phi_S : V \rightarrow W$  is also a linear isomorphism.

Hence,  $V \cong W$ .  $\square$

## 1.3 Example

### 1.3.1 Trivial Example

Let  $V$  be a vector space and let  $\text{id}_V : V \rightarrow V$  be the identity map on  $V$ . Then

*Solution.* Recall

$$\text{id}_V(\mathbf{v}) = \mathbf{v}, \quad \forall \mathbf{v} \in V.$$

So

1.  $\text{id}_V$  is a **linear map**.
2.  $\ker(\text{id}_V) = \{\mathbf{0}_V\}$ , hence  $\text{id}_V$  is **injective**.
3.  $\text{im}(\text{id}_V) = \underbrace{V}_{\text{entire domain}}$ , hence  $\text{id}_V$  is **surjective**.

Therefore,  $\text{id}_V : V \rightarrow V$  is a **linear isomorphism**. Consequently,  $V$  is isomorphic to itself.

Also, Every vector space is isomorphic to itself

□

## 2 Representations of Linear Maps as Matrices

In this section, we describe a systematic way to represent every linear map

$f : V \rightarrow W$  by a concrete matrix.

To do this, we begin a linear map  $f : V \rightarrow W$  where  $\dim_F(V) = n$  and  $\dim_F(W) = m$ .

Suppose we have fixed

$$B_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \text{ a basis of } V$$

$$B_2 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\} \text{ a basis of } W$$

Since  $f$  is linear, it is completely determined by its values on the basis vectors of  $V$ . For each  $\mathbf{v}_j \in B_1$ , we can write

$$f(\mathbf{v}_j) = \alpha_{1j}\mathbf{w}_1 + \alpha_{2j}\mathbf{w}_2 + \dots + \alpha_{mj}\mathbf{w}_m, \quad j = 1, \dots, n.$$

That is, we express each  $f(\mathbf{v}_j)$  as a linear combination of the basis vectors of  $W$ .

We now collect the coefficients into a matrix. Specifically, we define

$$[f]_{B_1}^{B_2} = (\alpha_{ij})_{i,j} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix},$$

whose  $j$ -th column is exactly the coordinate vector of  $f(\mathbf{v}_j)$  with respect to the basis  $B_2$ .

## 2.1 Definition

**Definition 5.2.1.** The matrix constructed above is called the *matrix representing  $f$  with respect to the bases  $B_1$  and  $B_2$* . We denote it by  $[f]_{B_1}^{B_2}$ .

**Definition 5.2.2.** Let  $A = (\alpha_{ij})$  be an  $l \times m$  matrix and  $B = \beta_{ij}$  be an  $m \times n$  matrix. Then the product  $AB$  is a  $l \times n$  matrix  $C = (\gamma_{ij})$  where,

$$(i, j)\text{-entry of } AB = \gamma_{ij} = \sum_{k=1}^m \alpha_{ik} \beta_{kj}$$

## 2.2 Theorem

**Theorem 5.2.1.** Let  $f : V \rightarrow W$  be a linear map between finite-dimensional vector spaces over a field  $F$ . Fix a basis  $B_1$  be a basis of  $V$ , and  $B_2$  be a basis of  $W$ .

Then, for every  $\mathbf{v} \in V$ , we have the equality of matrices (and vectors):

$$[f]_{B_1}^{B_2} [\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}.$$

*Proof.* Let  $B_1 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ ,  $B_2 = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$

Let

$$[f]_{B_1}^{B_2} = (\alpha_{ij})_{i,j} \iff f(\mathbf{v}_j) = \sum_{i=1}^m \alpha_{ij} \mathbf{w}_i, \forall j$$

And let

$$[\mathbf{v}]_{B_1} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \iff \mathbf{v} = \sum_{j=1}^n \lambda_j \mathbf{v}_j$$

Then

$$[f]_{B_1}^{B_2} [\mathbf{v}]_{B_1} = (\alpha_{ij})_{i,j} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^n \alpha_{1j} \lambda_j \\ \vdots \\ \sum_{j=1}^n \alpha_{mj} \lambda_j \end{pmatrix}.$$

On the other hand, since  $f$  is linear,

$$f(\mathbf{v}) = f\left(\sum_{j=1}^n \lambda_j \mathbf{v}_j\right) = \sum_{j=1}^n \lambda_j f(\mathbf{v}_j) = \sum_{j=1}^n \lambda_j \left(\sum_{i=1}^m \alpha_{ij} \mathbf{w}_i\right) = \sum_{i=1}^m \left(\sum_{j=1}^n \alpha_{ij} \lambda_j\right) \mathbf{w}_i.$$

Hence,

$$[f(\mathbf{v})]_{B_2} = \begin{pmatrix} \sum_{j=1}^n \alpha_{1j} \lambda_j \\ \vdots \\ \sum_{j=1}^n \alpha_{mj} \lambda_j \end{pmatrix},$$

which is exactly the vector obtained above. Therefore,

$$[f]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}.$$

*Remark.* The theorem can be summarized by the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \downarrow \phi_{B_1} & & \downarrow \phi_{B_2} \\ F^n & \xrightarrow{[f]_{B_1}^{B_2}} & F^m \end{array}$$

That is,

$$[f]_{B_1}^{B_2} \circ \phi_{B_1} = \phi_{B_2} \circ f,$$

or equivalently,

$$[f]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}, \quad \forall \mathbf{v} \in V.$$

□

□

## 2.3 Example

### 2.3.1 Matrix Representation of Differentiation

Consider the linear map  $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$  defined by  $p(x) \mapsto p'(x)$ . Fix the bases

$$S_1 = \{1, x, x^2, x^3\} \text{ of } \mathbb{R}[x]_{\leq 3}, \text{ and } S_2 = \{1, x, x^2\} \text{ of } \mathbb{R}[x]_{\leq 2}.$$

Find the matrix  $[D]_{S_1}^{S_2}$

*Solution.* We compute the images of the basis vectors of  $S_1$  and express them in terms of the basis  $S_2$ .

First,

$$D(\mathbf{1}) = 0 = \mathbf{0} \cdot 1 + \mathbf{0} \cdot x + \mathbf{0} \cdot x^2,$$

so

$$[D(\mathbf{1})]_{S_2} = (0, 0, 0)^T.$$

Next,

$$D(\mathbf{x}) = 1 = \mathbf{1} \cdot 1 + \mathbf{0} \cdot x + \mathbf{0} \cdot x^2,$$

so

$$[D(\mathbf{x})]_{S_2} = (1, 0, 0)^T.$$

Similarly,

$$D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2, \quad [D(x^2)]_{S_2} = (0, 2, 0)^T,$$

and

$$D(x^3) = 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2, \quad [D(x^3)]_{S_2} = (0, 0, 3)^T.$$

Placing these coordinate vectors as columns, we obtain

$$[D]_{S_1}^{S_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

*Remark.* For example, if

$$p(x) = 1 + 2x + 4x^3, \quad D[p(x)] = 2 + 12x^2$$

then

$$[p(x)]_{S_1} = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 4 \end{pmatrix}, \quad [D(p(x))]_{S_2} = \begin{pmatrix} 2 \\ 0 \\ 12 \end{pmatrix}.$$

We observe that

$$[D(p(x))]_{S_2} = [D]_{S_1}^{S_2} [p(x)]_{S_1}.$$

Thus differentiation has been translated into matrix multiplication. □

□

### 2.3.2 Matrix Representations under Different Bases

Consider the linear map  $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ , defined by

$$(x, y, z) \mapsto (-x - 2y + 2z, 4x + 3y - 4z, -2y + z).$$

- (a) Verify that  $f$  is a linear transformation.
- (b) Find the matrix representing  $f$  with respect to the standard basis of  $\mathbb{R}^3$  (for both the domain and codomain).
- (c) Verify that

$$S = \{(1, 0, 1), (0, 1, 1), (1, -1, 1)\}$$

is a basis of  $\mathbb{R}^3$ . Find the matrix  $[f]_S^S$ .

*Solution.* The following are the solutions of three subproblems

(a)

For any  $\mathbf{u} = (x_1, y_1, z_1)$ ,  $\mathbf{v} = (x_2, y_2, z_2) \in \mathbb{R}^3$ ,

$$\begin{aligned} f(\mathbf{u} + \mathbf{v}) &= f(x_1 + x_2, y_1 + y_2, z_1 + z_2) \\ &= \left( -(x_1 + x_2) - 2(y_1 + y_2) + 2(z_1 + z_2), \right. \\ &\quad \left. 4(x_1 + x_2) + 3(y_1 + y_2) - 4(z_1 + z_2), \right. \\ &\quad \left. -2(y_1 + y_2) + (z_1 + z_2) \right) \\ &= f(\mathbf{u}) + f(\mathbf{v}). \end{aligned}$$

Moreover, for any  $\lambda \in \mathbb{R}$ ,

$$\begin{aligned} f(\lambda \mathbf{u}) &= f(\lambda x_1, \lambda y_1, \lambda z_1) \\ &= \left( -\lambda x_1 - 2\lambda y_1 + 2\lambda z_1, 4\lambda x_1 + 3\lambda y_1 - 4\lambda z_1, -2\lambda y_1 + \lambda z_1 \right) \\ &= \lambda f(\mathbf{u}). \end{aligned}$$

Therefore,  $f$  satisfies both additivity and homogeneity. Hence,  $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  is a linear transformation.

(b) Let  $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$  be the standard basis of  $\mathbb{R}^3$ .

Compute the images of the basis vectors:

$$\begin{aligned} f(\mathbf{e}_1) &= (-1, 4, 0) = (-1) \mathbf{e}_1 + 4 \mathbf{e}_2 + 0 \mathbf{e}_3, \\ f(\mathbf{e}_2) &= (-2, 3, -2) = (-2) \mathbf{e}_1 + 3 \mathbf{e}_2 + (-2) \mathbf{e}_3, \\ f(\mathbf{e}_3) &= (2, -4, 1) = 1 \mathbf{e}_1 + (-4) \mathbf{e}_2 + 1 \mathbf{e}_3. \end{aligned}$$

Hence,

$$[f]_B^B = \begin{pmatrix} -1 & -2 & 2 \\ 4 & 3 & -4 \\ 0 & -2 & 1 \end{pmatrix}.$$

(c)

Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1, 0, 1), (0, 1, 1), (1, -1, 1)\}.$$

Compute

$$\begin{aligned} f(\mathbf{v}_1) &= (1, 0, 1) = 1\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3, \\ f(\mathbf{v}_2) &= (0, -1, -1) = 0\mathbf{v}_1 + (-1)\mathbf{v}_2 + 0\mathbf{v}_3, \\ f(\mathbf{v}_3) &= (3, -3, 3) = 0\mathbf{v}_1 + 0\mathbf{v}_2 + 3\mathbf{v}_3. \end{aligned}$$

Therefore,

$$[f]_S^S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

*Remark.* Although the **moral standard basis is convenient** for computation, **it is not necessarily the "nicest" basis for representing a linear transformation.**

Question: How to find the "nice" basis for a linear map  $f$ ?

Answer: Diagonalization Theory

□

□

### 2.3.3 Using Matrix Representations to Study a Linear Map

Consider the function  $f : \mathbb{R}[x]_{\leq 2} \longrightarrow \mathbb{R}[x]_{\leq 3}$

$$p(x) \longmapsto -\frac{d}{dx}(p(x^2 - 1)) + \int_2^x p(t) dt.$$

- (a) Verify that  $f$  is a linear transformation.
- (b) Find  $\ker(f)$ . Is  $f$  injective?
- (c) Find  $\text{rank}(f)$ . Is  $f$  surjective?

*Solution.* The following are the solutions of above three subproblems

(a)

Let  $p(x), q(x) \in \mathbb{R}[x]_{\leq 2}$  and let  $\lambda \in \mathbb{R}$ .

Observe that differentiation, composition with  $x^2 - 1$ , and integration are all linear operators. Hence

$$\begin{aligned} f(p + q) &= -\frac{d}{dx}((p + q)(x^2 - 1)) + \int_2^x (p(t) + q(t)) dt \\ &= -\frac{d}{dx}(p(x^2 - 1)) - \frac{d}{dx}(q(x^2 - 1)) \\ &\quad + \int_2^x p(t) dt + \int_2^x q(t) dt \\ &= f(p) + f(q). \end{aligned}$$

Similarly,

$$f(\lambda p) = -\frac{d}{dx}(\lambda p(x^2 - 1)) + \int_2^x \lambda p(t) dt = \lambda \left( -\frac{d}{dx}(p(x^2 - 1)) + \int_2^x p(t) dt \right) = \lambda f(p).$$

Therefore,  $f$  satisfies additivity and homogeneity. Hence,  $f$  is a linear transformation.

(b)

Fix a basis  $S_1 = \{1, x, x^2\}$  (domain), and a basis  $S_2 = \{1, x, x^2, x^3\}$  (co-domain).

Compute the images of the basis vectors:

$$f(1) = x - 2 = (-2) + 1x + 0x^2 + 0x^3$$

$$f(x) = -2x + \frac{1}{2}x^2 - 2 = (-2) + (-2)x + \frac{1}{2}x^2 + \left(\frac{-11}{3}\right)x^3$$

$$f(x^2) = -\frac{1}{3}x^3 + 4x - \frac{8}{3} = \left(\frac{-8}{3}\right) + 4x + 0x^2 + \left(\frac{-11}{3}\right)x^3$$

Hence

$$A := [f]_{S_1}^{S_2} = \begin{pmatrix} -2 & -2 & \frac{-8}{3} \\ 1 & -2 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{-11}{3} \end{pmatrix}.$$

Since

$$\ker(f) \cong \ker([f]_{S_1}^{S_2}),$$

and the above matrix has trivial nullspace,

$$\ker(A) = \ker([f]_{S_1}^{S_2}) = \{\mathbf{0}\} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

we conclude that

$$\ker(f) = \{0 \text{ (zero polynomial)}\},$$

where 0 denotes the zero polynomial.

Therefore,  $f$  is injective.

(c)

By the Rank–Nullity Theorem,

$$\text{rank}(f) + \text{nullity}(f) = \dim(\mathbb{R}[x]_{\leq 2}).$$

Since  $\ker(f) = \{0\}$ , we have  $\text{nullity}(f) = 0 \implies \text{rank}(f) = 3$ , however,  $\dim(\mathbb{R}[x]_{\leq 3}) = 4$

$$\implies \text{rank}(f) \neq \dim(\underbrace{\mathbb{R}[x]_{\leq 3}}_{\text{co-domain}})$$

$\implies f$  is NOT surjective.

□

### 3 Composition of linear maps and matrix multiplication

#### 3.1 Definition

**Definition 5.3.1.** Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$  be two functions. The *composition* of  $f$  and  $g$  is the function  $g \circ f : A \rightarrow C$  is defined by

$$(g \circ f)(a) = g(f(a)), \quad \forall a \in A.$$

For  $f : A \rightarrow A$  then we denote  $f^n := \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}$ .

#### 3.2 Theorem

**Theorem 5.3.1.** Let  $f : V \rightarrow U$  and  $g : U \rightarrow W$  be linear maps between finite-dimensional vector spaces over a field  $F$ . Fix

a basis  $B_1$  of  $V$ , a basis  $B_2$  of  $U$ , a basis  $B_3$  of  $W$

Then we have an equality of matrices:

$$[g \circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}.$$

*Proof.* Let

$B_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$  be a basis of  $V$

$B_2 = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$  be a basis of  $U$

$B_3 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p\}$  be a basis of  $W$

Let

$$[f]_{B_1}^{B_2} = (\alpha_{ij})_{i,j} \text{ which means } f(\mathbf{v}_j) = \sum_{k=1}^n \alpha_{kj} \mathbf{u}_k$$

$$[g]_{B_2}^{B_3} = (\beta_{ik})_{i,k} \text{ which means } g(\mathbf{u}_k) = \sum_{i=1}^p \beta_{ik} \mathbf{w}_i$$

We now determine the matrix

$$[g \circ f]_{B_1}^{B_3}.$$

For each basis vector  $\mathbf{v}_j$ ,

$$\begin{aligned}
(g \circ f)(\mathbf{v}_j) &= g(f(\mathbf{v}_j)) \\
&= g\left(\sum_{k=1}^n \alpha_{kj} \mathbf{u}_k\right) \\
&= \sum_{k=1}^n \alpha_{kj} g(\mathbf{u}_k) \quad (\text{since } g \text{ is linear}) \\
&= \sum_{k=1}^n \alpha_{kj} \sum_{i=1}^p \beta_{ik} \mathbf{w}_i \\
&= \sum_{i=1}^p \left( \sum_{k=1}^n \beta_{ik} \alpha_{kj} \right) \mathbf{w}_i.
\end{aligned}$$

Hence, the  $(i, j)$ -entry of  $[g \circ f]_{B_1}^{B_3}$  is  $\sum_{k=1}^n \beta_{ik} \alpha_{kj}$ , which is exactly the  $(i, j)$ -entry of the matrix product  $[g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}$ .

Therefore,

$$[g \circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}.$$

□

### 3.3 Example

#### 3.3.1 Composition of Matrix Multiplication

The point  $(a, b)$  in  $\mathbb{R}^2$  is first rotated counterclockwise by  $\frac{\pi}{4}$  about the origin and then reflected about the line  $y = \sqrt{3}x$ . Find the resulting point.

*Solution.* Let

$B$  be the standard basis of  $\mathbb{R}^2$

$T_1$  : rotation by  $\frac{\pi}{4}$  (anti-clockwise)

$T_2$  : reflection along the line  $y = \tan\left(\frac{\pi}{3}\right)x$

Then

$$\begin{aligned}
[T_1]_B^B &= \begin{pmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \\
[T_2]_B^B &= \begin{pmatrix} \cos \frac{2\pi}{3} & \sin \frac{2\pi}{3} \\ \sin \frac{2\pi}{3} & -\cos \frac{2\pi}{3} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.
\end{aligned}$$

Hence,

$$\begin{aligned}
[T_2 \circ T_1]_B^B &= [T_2]_B^B \cdot [T_1]_B^B \text{ (matrix multiplication)} \\
&= \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\
&= \begin{pmatrix} \frac{-1 + \sqrt{3}}{2\sqrt{2}} & \frac{1 + \sqrt{3}}{2\sqrt{2}} \\ \frac{1 + \sqrt{3}}{2\sqrt{2}} & \frac{1 - \sqrt{3}}{2\sqrt{2}} \end{pmatrix}.
\end{aligned}$$

Therefore,

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} \frac{-1 + \sqrt{3}}{2\sqrt{2}} & \frac{1 + \sqrt{3}}{2\sqrt{2}} \\ \frac{1 + \sqrt{3}}{2\sqrt{2}} & \frac{1 - \sqrt{3}}{2\sqrt{2}} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{-1 + \sqrt{3}}{2\sqrt{2}}a + \frac{1 + \sqrt{3}}{2\sqrt{2}}b \\ \frac{1 + \sqrt{3}}{2\sqrt{2}}a + \frac{1 - \sqrt{3}}{2\sqrt{2}}b \end{pmatrix}}_{\text{in standard basis of } B}.$$

□

## 4 Inverse of a matrix

### 4.1 Definition

**Definition 5.4.1.** Let  $A \in M_n(F)$  be a square matrix. If there exists a matrix  $B \in M_n(F)$  s.t.  $AB = BA = I_n$ , then

1. we say that  $A$  is *invertible* (or *non-singular*); and
2. we call  $B$  the *inverse* of  $A$  and denote it by  $A^{-1}$ .

### 4.2 Theorem

**Theorem 5.4.1.** Let  $A \in M_{m \times n}(F)$ . Suppose there exists  $B \in M_{n \times m}(F)$  s.t.

$$BA = I_n \text{ and } AB = I_m.$$

Then  $n = m = \text{rank}(A)$ .

*Proof.* We shall show that  $\text{rank}(A) = m = n$ .

(1) First observe that

$$\text{rank}(A) = \text{rank}(A \overbrace{BA}^{I_n}) \leq \text{rank}(AB) = \text{rank}(I_m) = m,$$

On the other hand,

$$m = \text{rank}(I_m) = \text{rank}(AB) \leq \text{rank}(A).$$

Therefore,

$$\text{rank}(A) = m.$$

(2) Similarly,

$$\text{rank}(A) = \text{rank}(ABA) \leq \text{rank}(BA) = \text{rank}(I_n) = n,$$

Also,

$$n = \text{rank}(I_n) = \text{rank}(BA) \leq \text{rank}(A).$$

Hence,

$$\text{rank}(A) = n.$$

Therefore, if  $A$  has an inverse, then

$$n = m = \text{rank}(A).$$

*Remark.* Rank inequalities

$$\begin{cases} \text{rank}(f \circ g) \leq \text{rank}(f) \\ \text{rank}(f \circ g) \leq \text{rank}(g) \end{cases}$$

and **matrix version**

$$\begin{cases} \text{rank}(AB) \leq \text{rank}(A) \\ \text{rank}(AB) \leq \text{rank}(B) \end{cases}$$

□

□

**Theorem 5.4.2.** Let  $f : V \rightarrow W$  be an isomorphism of finite-dimensional vector spaces. Fix any basis  $B_1$  of  $V$  and  $B_2$  of  $W$ . Then we have an equality of matrices

$$([f]_{B_1}^{B_2})^{-1} = [f^{-1}]_{B_2}^{B_1}.$$

In other words, the inverse of a matrix represents the inverse of a linear map.

*Proof.* By Theorem 5.3.1,

$$[f^{-1}]_{B_2}^{B_1} [f]_{B_1}^{B_2} = [f^{-1} \circ f]_{B_1}^{B_1}.$$

Since  $f^{-1} \circ f = \text{id}_V$ , we obtain

$$[f^{-1} \circ f]_{B_1}^{B_1} = [\text{id}_V]_{B_1}^{B_1} = I_n, \text{ where } n = \dim(V)$$

Therefore,

$$[f^{-1}]_{B_2}^{B_1} = ([f]_{B_1}^{B_2})^{-1}.$$

□

## 5 Remark

| Vector space / Linear map world        | Vector / Matrix world                                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| $f : V \rightarrow W$                  | $[f]_{B_1}^{B_2}$ ( <b>matrix</b> )                                                                                              |
| Composition $f \circ g$                | Multiplication $[f \circ g]_{B_1}^{B_3} = [f]_{B_2}^{B_3} [g]_{B_1}^{B_2}$                                                       |
| Addition $f + g$ , Scaling $\lambda f$ | Addition $[f + g]_{B_1}^{B_2} = [f]_{B_1}^{B_2} + [g]_{B_1}^{B_2}$ , Scaling $[\lambda f]_{B_1}^{B_2} = \lambda [f]_{B_1}^{B_2}$ |
| Inverse $f^{-1} : W \rightarrow V$     | Matrix inverse $[f^{-1}]_{B_2}^{B_1} = ([f]_{B_1}^{B_2})^{-1}$                                                                   |

Question: When is a matrix invertible?